Executable Evidence Management for CAD-Guided Aircraft Inspection

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Abstract

An inspection quality statement can become inapplicable when a camera, image-processing component or engineering requirement changes, even though part of the numerical result remains usable. We present an executable evidence interface for maintaining CAD-guided inspection services through such changes. Typed requests compile to scene-level quality events, and calibrated bounds transfer through event containment and compatible producer functions. Exact-event fallback preserves the services admitted by direct renewal, while versioned repair transactions restore the association between evidence and the current authenticated output. A sequential experiment uses 256 calibration scenes and 64 independent test scenes, with each test scene undergoing 16 operations including combined updates, stale delivery, repair and numerical rollback. Full-family comparisons support the same 54 of 64 calibrated contexts. Containment reduces full-catalogue record evaluations from 13,824 to 8,960 (35.19%). All complete implementations restore their 128 eligible repair checkpoints without issuing an invalid binding. Dependency caching separately reduces measured workflow CPU by 59.31%. A complementary light-transport challenge evaluates 256 images from 64 independent surface scenes spanning four structural families and four imaging conditions. The original and development-selected extractors violate the association event on 82 and 104 images; the observable screen rejects every observed quality failure. The results connect evidence renewal and output recovery to explicit inspection requirements, while distinguishing numerical reuse, statistical applicability and geometric quality.

Keywords: Engineering knowledge representation, Evidence reuse, Engineering change management, Risk calibration, Aircraft inspection

1. Introduction

An aircraft inspection image becomes useful when its observations can be located on the airframe and assessed against an engineering requirement. CAD-guided aircraft inspection establishes this spatial correspondence from model features [1]; fastener-inspection systems then evaluate surface components under the available imaging conditions [2]. During a repeated camera pass, increasing exposure can recover surface contrast while increasing motion blur. The airframe geometry remains unchanged, yet the processing configuration may associate a different set of fasteners with the CAD model. A quality statement established for the earlier configuration must therefore be checked before it accompanies the new output.

Such decisions accumulate within a continuing inspection workflow. A relaxed mapping tolerance may make an existing result usable, while a delayed output can arrive after its processing configuration has been replaced. Requirement-based digital-twin studies identify context, uncertainty, thresholds and historical information as inputs to maintenance decisions [3]. Selective sensor calibration concentrates effort on measurements identified as unreliable [4]. For inspection, the corresponding problem is to retain computations and quality statements that still apply, renew those affected by a change, and attach the result to the correct observation. The decision must survive successive updates and repairs, including a legitimate return to an earlier numerical configuration.

Existing research supplies substantial foundations for this process. CAD consistency analysis identifies repairs after model edits [5], lifecycle knowledge graphs preserve specification, version and operating-state histories [6], and dynamic safety cases update evidence when monitored conditions change [7]. Provenance

records describe how an output was generated and revised [8]. We study how these records support a quantitative inspection decision: whether a calibrated event covers the requested quality, whether the functions defining that event remain compatible, and whether the evidence is attached to the authenticated current output.

We develop an executable evidence interface that carries an inspection service through these changes. First, typed requests compile to quality events whose calibrated bounds can be reused by event containment; exact-event fallback preserves the services supported by direct renewal under the same calibration. Second, a versioned repair transaction restores the association to an authenticated output while retaining its change history and the event's admission decision. Third, a sequential change campaign evaluates these decisions against dependency-aware renewal and independent registry implementations, separating numerical computation from evidence preparation. A complementary image challenge uses four surface structures and four imaging conditions to examine the quality event and its observable screening signals. Together, the experiments connect selective renewal to the inspection service that can resume after each change.

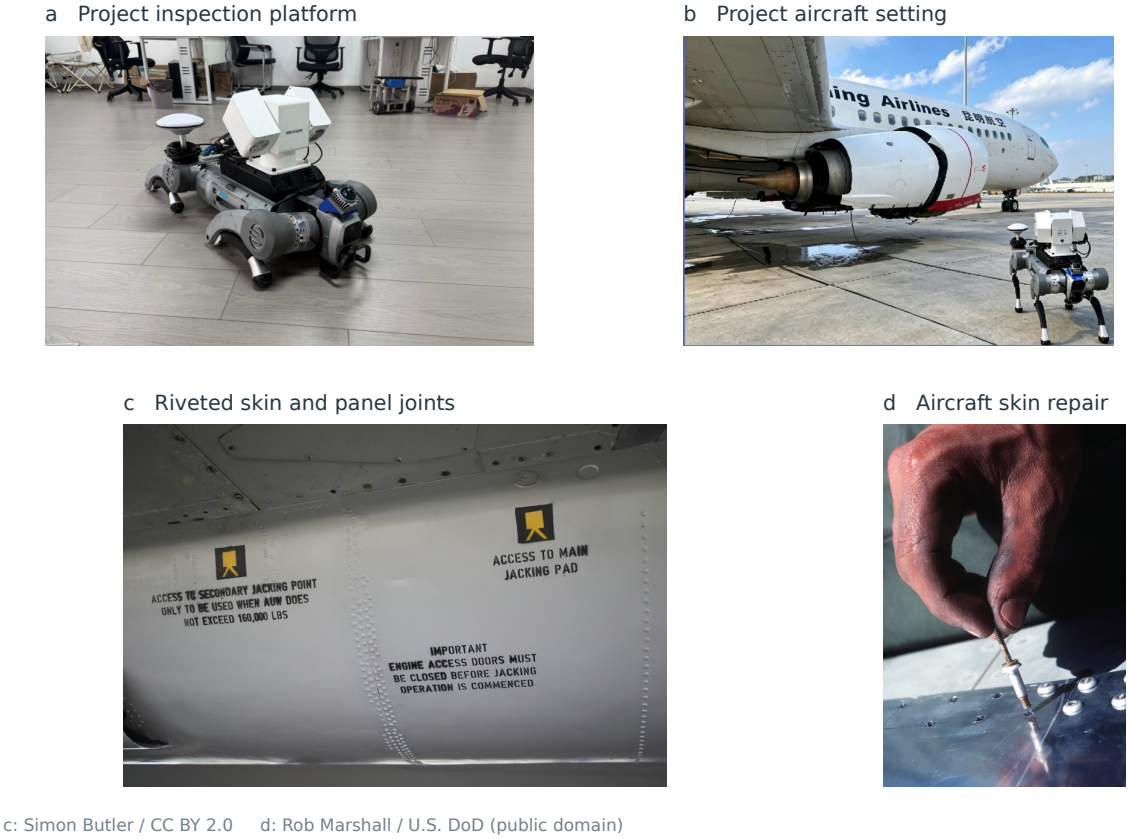


Figure 1: Aircraft inspection at platform and surface scales. (a,b) Original project photographs. (c) Under-fuselage panels on Avro Vulcan XM655, photographed by Simon Butler (CC BY 2.0). (d) Aircraft sheet-metal repair, photographed by Rob Marshall (U.S. Department of Defense, public domain). Sources and image credits are archived in the Supplementary Material.

2. Related work

2.1. Engineering knowledge and change-dependent decisions

Engineering knowledge representations connect requirements, model entities and decisions. Petnga and Austin integrate ontological reasoning with physical quantities and unit-aware computations [9]. Zou and

Feng resolve inconsistencies introduced by direct CAD editing through constraint analysis and ranked repair candidates [5]. These approaches establish the role of an actionable consistency decision. Quinn and McArthur assess representational completeness and expressiveness through industrial concept mappings and application relations [10]; inspection-robot ontologies relate task requirements to hardware capabilities [11].

Recent AEI systems connect engineering representations to executable models and lifecycle histories. Xiao et al. map design and building-model information into executable energy-analysis models [12]. Yang et al. separate persistent entities, specifications, released versions and runtime states, with explicit change actions and retained histories [6]. Our interface adopts this distinction between identity and changing state. Its maintained object is a quantitative quality event: the event's units, thresholds and producer dependencies determine which calibrated statement supports a request.

2.2. Provenance, selective recomputation and evolving evidence

The W3C PROV model describes generation, use, derivation, revision and invalidation; its constraints formalize consistency among these records [8, 13]. PDIFF compares execution provenance to locate differences caused by data, service or workflow revisions [14]. RECOMP studies selective recomputation as data and methods evolve [15], and version-aware restart trees identify affected outputs for recomputation [16]. Build-system theory supplies dependency-directed retention of unchanged computations [17]. These results motivate the dependency cache used by all matched incremental methods in this study.

Evidence maintenance also has an established assurance literature. Dynamic safety cases couple monitored variables, confidence structures and update rules, including removal of arguments whose assumptions become invalid [7]. SafeMachine organizes machine-safety assessment around traceable, context-aware evidence [18]. Assumption–guarantee contracts specify component obligations [19], and runtime-assurance architectures mediate permissible execution [20]. The present interface specializes evidence maintenance to a calibrated scene-quality event and the output receiving that statement. Its recovery transaction checks the current trusted binding; its event rule determines whether the requested service can resume.

2.3. Risk calibration and visual inspection

Learn then Test selects predictive programs using hypothesis tests with high-probability risk control [21]. Conformal Risk Control calibrates a monotone loss to control its expectation [22]. We use exact binomial inversion [23] with a fixed finite-family allocation. Event inclusion supplies an ordering for evidence reuse, and exact fallback reserves the original request's test. The resulting risk statement has a specified population, producer and quality event.

Industrial visual inspection also requires the event to reflect the task. Component-aware anomaly detection separates component structure and logical arrangement from local appearance [24]. CAD-guided aircraft inspection and rivet inspection provide the spatial and component-level setting for our capture-to-CAD requirements [1, 2]. Appearance-randomized rendering can improve transfer across image domains [25]; Xu et al. evaluate simulation-to-experiment adaptation for ultrasonic concrete-damage signals [26]. Our geometric image challenge examines extraction, correspondence and mapping across surface structures and imaging conditions. Its scene population is recorded separately from the population used to calibrate the service.

Table 1: Closest research connections and the inspection decision examined here.

Research connection	Established capability	Inspection decision
Lifecycle knowledge [6]	Preserve identities, versions, states and change dependencies	Which requested quantities remain covered after a producer change?
Dynamic evidence [7]	Update arguments and evidence as monitored conditions change	Which quality event has an applicable numerical risk bound?
Workflow provenance [14, 16]	Locate changed computations and affected outputs	Which statements can accompany retained or corrected outputs?
Risk-controlled selection [21]	Admit programs under a calibrated risk allowance	How can reuse preserve exact-renewal support?

3. Inspection service and evidence model

3.1. A continuing inspection decision

Consider a camera observation used to associate fastener positions with CAD addresses. A complete-service request constrains spatial sampling, motion blur, correct-address coverage, fitted residual and physical mapping error. Increasing exposure preserves the geometric sampling scale but changes blur and image formation. A combined update to exposure, illumination and extraction changes several functions at once. The evidence interface determines which earlier statements remain applicable and constructs an exact-event statement when a compatible stronger one is unavailable.

The output also carries an observation identity, producer signature and registry revision. An old output delivered after an update fails the current binding check. Recovery then verifies the trusted output and commits a newer revision. A subsequent rollback can restore an earlier numerical configuration while advancing the registry revision again. Figure 2 shows the relation between image quality and statistical admission for the first test source; the sequential experiment then follows its output through repair and rollback. The first held-out observation illustrates this distinction (Fig. 2). Its sampling scale remains 0.658 mm/pixel after the exposure update, while blur increases from 0.776 to 1.163 pixels. Both individual outputs satisfy the strict complete-service predicate. The independently calibrated upper bound nevertheless changes from 0.093 to 0.292, above the allowance 0.15. The relaxed complete event has upper bound 0.104 and remains supported. The workflow therefore continues the relaxed service while withholding the strict one; the output's current producer and binding determine which statement accompanies it.

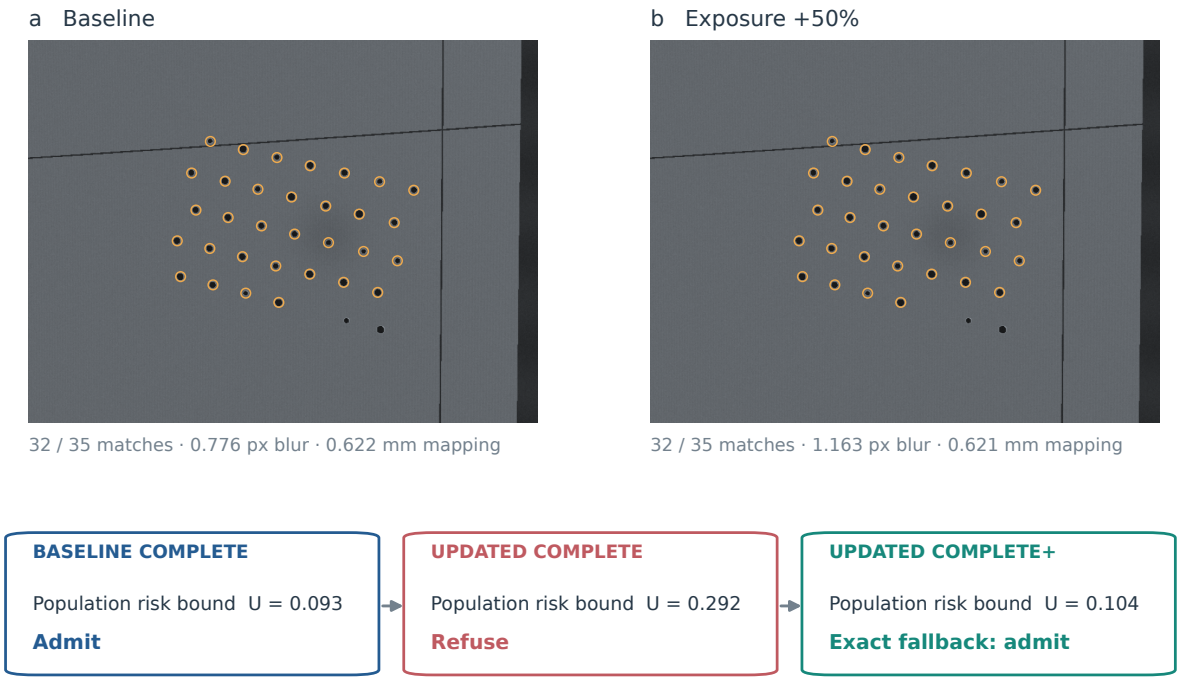


Figure 2: Actual controlled-raster inputs from the first held-out source before and after an exposure update. Orange rings mark centers extracted from pixels. Both particular images satisfy the strict quality event, while the independently calibrated population bounds support different service decisions. Exact fallback retains the relaxed complete service after the update.

3.2. Typed requests, events and dependencies

An engineering request is $R = (g, a, \varphi, \delta)$: population g, service a, quality predicate φ and allowable scene-level failure probability δ. Its grammar is

$$\varphi ::= \mathrm{Interval}(m, u, \ell, h) \mid \varphi \wedge \varphi, \tag{1}$$

where m identifies a quantity and u its engineering unit. Following unit-aware engineering knowledge models [9], compilation checks dimensions, converts units, intersects repeated bounds and canonicalizes the event. Nonfinite required values fail the quality predicate; an absent required field prevents evaluation as a schema error. The accepted set and its failure event are

$$S_R = \{z : \ell_{R,j} \le z_j \le h_{R,j},\ j \in J_R\}, \qquad E_{p,R} = \{z_p(X) \notin S_R\}, \tag{2}$$

where p identifies the numerical producer and X is a complete scene from population g.

Typed records connect the request to its evidence and output (Table 2). Their relation signatures include requires : Request $\to$ Event, bounds : Card $\to$ Event, dependsOn : Quantity $\to$ Provider and uses : Receipt $\to$ SourceVersion. In the schema, Receipt denotes the execution record. Schema validation checks relation endpoints; request compilation resolves the event's quantities and dependencies. The representation stores immutable specifications separately from active registry revisions.

For quantity j, the provider manifest declares dependencies D_j; the request uses $D_R = \bigcup_{j \in J_R} D_j$. Its producer projection identifies the corresponding registered functions and parameters. Within each source, a numerical stage also uses an ordinary content-addressed key [17],

$$h_v = \mathrm{Hash}\big(v, \sigma_v, \{h_u : u \in \mathrm{pa}(v)\}\big), \tag{3}$$

where σ_v identifies the stage implementation and settings. A cached value is retained when the registered stage and its inputs agree. The experiments fix implementation code within each run and maintain separate numerical caches for different sources. This stage cache controls numerical recomputation. The request projection separately selects the functions relevant to the quality statement. Before a component update is registered, a paired regression gate re-executes common inputs and rejects a declared invariance if an unaffected quantity changes. Runtime decisions use the registered signatures.

Table 2: Records connecting a quality requirement to its current output.

Record	Required content	Decision
Request	Population, service, units, intervals, allowance	Compile event and dependencies
Evidence card	Event, producer, calibration lineage, n, k, U	Cover the requested event
Source version	Observation identity, active producer, output digest	Authenticate the current binding
Repair transaction	Expected revision, trusted output and new revision	Restore a quarantined binding
Execution record	Plan, attachment and transaction history	Explain issuance or refusal

3.3. Image-to-CAD quality

Pixel extraction supplies candidate fastener centers. A pose prior projects nominal CAD locations under the perspective camera model [27], and minimum-cost assignment uses image-plane distances within a prior gate. The implementation uses the modified Jonker–Volgenant assignment routine described by Crouse [28]. For assigned correspondences $\mathcal{I}$, pose refinement solves

$$\begin{aligned}\widehat{\xi} = \arg\min_{\xi} \sum_{i \in \mathcal{I}} \sum_{d \in \{x,y\}} \rho\left(\frac{\pi_\xi(q_i)_d - u_{i,d}}{\sigma_u}\right) \\ + \sum_{d=1}^{3} \left[\rho\left(\frac{\Delta t_d}{\sigma_t}\right) + \rho\left(\frac{\Delta \theta_d}{\sigma_\theta}\right)\right].\end{aligned} \tag{4}$$

Here q_i is a nominal CAD address, u_i a pixel-derived center, and π_ξ the perspective projection. Translation and local rotation perturb the prior. The componentwise Huber loss [29] has transition 2. Estimation uses pixels, nominal CAD and the pose prior.

Offline scoring evaluates the correspondence and physical map after estimation. Correct-address coverage is the fraction of planned sites assigned to their physical counterparts. Fitted residual includes every assigned correspondence, including incorrect assignments. Maximum mapping error evaluates the estimated image-to-CAD map $H_{\widehat{\xi}}$ at every planned physical image location:

$$e_{\mathrm{CAD}} = \max_{i \in \mathcal{P}} \| H_{\widehat{\xi}}(u_i^\star) - q_i^\star \|. \tag{5}$$

Scene geometry supplies these evaluation labels after estimation. In the geometric challenge, the map intersects the estimated camera ray with the exterior skin CAD mesh. Planned sites remain in the evaluation under occlusion, and a missing intersection gives infinite error. The observable image screen uses matched-site count and fitted residual; calibration and evaluation use every requested quality quantity, including correct-address coverage and mapping error.

4. Service maintenance through change

4.1. Validity of calibration-selected services

For k failures among n independent complete calibration scenes, a one-sided Clopper–Pearson bound [23] is

$$U(k, n; \alpha) = \begin{cases} 1, & k = n, \\ \mathrm{Beta}^{-1}(1 - \alpha; k + 1, n - k), & k < n. \end{cases} \tag{6}$$

Fix a family of K producer/event functions before calibration and allocate one-sided error levels whose sum is at most β. The common confidence event then covers all K risks, using the familywise selection principle of Learn then Test [21]. The experiments use equal allocation $\alpha = \beta/K$. A service plan selected from these bounds is committed before evaluating a new scene.

A stronger requirement provides evidence for a weaker one: $S_A \subseteq S_R$ implies $E_{p,R} \subseteq E_{p,A}$. Producer transfer additionally requires identity of the requested functions,

$$z_{p',j}(x) = z_{p,j}(x) \quad \text{for all } j \in J_R \text{ and supported } x. \tag{7}$$

Registered dependency projections express this interface condition. The statistical guarantee is conditional on functional identity; the component regression gate tests the declared implementation on archived inputs.

Proposition 1 (Validity of calibration-selected services)**.** *Let $C \sim P^n$ be calibration data and let a new scene $X \sim P$ be independent of C. Fix K producer/event functions with one-sided allocations summing to at most β. With probability at least $1 - \beta$ over C, every admitted request selected using C has fresh-scene risk at most δ, provided that its event is pointwise contained in an anchor event from this fixed family with $U_A(C) \le \delta$ and its transferred functions satisfy Eq. (7).*

Proof. The union bound gives a calibration event of probability at least $1 - \beta$ on which every fixed-family risk is bounded by its $U_A(C)$. On that event, pointwise inclusion gives $\Pr_X(E_R \mid C) \le \Pr_X(E_A \mid C) \le U_A(C) \le \delta$ for every eligible selection. Requested-function identity preserves the inclusion after transfer. □

The proposition concerns the quality of a planned service on a fresh scene. For a subsequent issuance indicator I, it also gives $\Pr(E_R \cap \{I = 1\} \mid C) \le \delta$ on the same calibration event. The conditional failure rate among outputs accepted by a scene-dependent screen, $\Pr(E_R \mid I = 1, C)$, is a separate quantity. We measure screen acceptance and accepted-but-failed outcomes explicitly in the image challenge.

4.2. Containment with exact-event fallback

For a committed producer, the interface first searches compatible covering cards with $U_A \le \delta$. If none qualifies, it constructs the request's exact-event card from the fixed family. The route is

$$Q(R) = \begin{cases} \text{reuse}(A), & \mathcal{A}(R) \neq \varnothing, \\ \text{exact}(R), & \mathcal{A}(R) = \varnothing,\ U_R \le \delta, \\ \text{refuse}, & \text{otherwise.} \end{cases} \tag{8}$$

Here $\mathcal{A}(R)$ contains cards with matching population and service, compatible producer projection, and a stronger accepted set whose bound satisfies $U_A \le \delta$. Unsuccessful anchors and refusal cards count toward construction work.

Proposition 2 (Support preservation with exact fallback)**.** *Suppose the compared cards use the same n calibration scenes and allocation α, and transferred request quantities agree on those scenes. Containment followed by exact-event fallback admits exactly the requests admitted by the complete exact-event catalogue.*

Proof. Containment gives $k_R \le k_A$. At fixed n and α, the binomial upper bound is nondecreasing in k, so admission by an anchor implies $U_R \le U_A \le \delta$. Otherwise, fallback evaluates the same exact event as the complete catalogue. □

The fixed family reserves an allocation for every exact fallback. A failed strict anchor can therefore lead to an admissible relaxed event within the original confidence budget. A change in registered population identity selects a different evidence population before any transfer decision.

4.3. Quarantine, repair and revision order

Provenance distinguishes the generation, revision and invalidation of an output [8, 13]. The output binding identifies a source, producer, version and payload digest. An authoritative registry stores the current tuple. Issuance requires both an admissible plan and agreement with that tuple. An invalid attachment places its binding workflow in quarantine and records the rejected attempt. Ordinary retries preserve the quarantine.

Recovery uses an explicit transaction with the expected registry revision, the trusted active output and the next registry revision. The registry verifies the producer, payload digest, source lineage and calibration identity before accepting the transaction. A successful repair replaces the binding record and retains its history. A missing repair certificate, wrong digest, replayed transaction or unregistered alias is rejected. An invalid command preserves any previously valid committed state.

The resulting invariant is that every issued binding equals the current authenticated registry record, and every transition out of quarantine has a successful repair transaction in its history. Revision order belongs to this control history. Restoring an earlier numerical configuration therefore creates a new revision; it does not authorize a late attachment carrying the old revision. The event card continues to determine whether that producer supports the current request after repair or rollback.

5. Experimental design

5.1. Scene population and calibrated requirements

The lifecycle experiment samples 256 independent calibration scenes and 64 independent test scenes from a controlled image population. Each scene draws its own surface curvature, camera pose, material appearance and physical-to-CAD displacements. Four fastener layouts are represented: parallel rows, curved rows, staggered sites and an access-panel perimeter. The layouts contain 28–36 sites, with fastener radii of 2.0–3.4 mm and typical spacing of 28–40 mm. Shared and local displacement standard deviations are 0.4 and 0.15 mm; site missingness is 0.025 and edge occlusion occupies 0–10% of the image.

Baseline images have 768×576 pixels, focal length 1,200–1,600 pixels, camera distance 0.70–1.15 m, exposure 3–7 ms and surface-relative motion 0.06–0.18 m/s. The family contains the baseline and five

component updates: focal length $\times 1.15$, exposure $\times 1.5$, image region 896×672, illumination $\times 0.45$ and extraction contrast threshold $18 \to 10$. Two combined producers apply exposure/illumination/extraction together and focal length/image region together. Named configurations are complete settings; only the two explicitly combined producers apply simultaneous changes.

All eight producers execute on every calibration scene, yielding 2,048 outputs. The eight requests in Table 3 give the fixed family $K = 8 \times 8 = 64$, with $\beta = 0.05$, $\alpha = 0.05/64$ and primary allowance $\delta = 0.15$. The allowance concerns violation of the capture-to-CAD quality conjunction by a complete scene. Calibration selects the plans before the test inputs are processed. The six producers visited by the sequential campaign execute on all 64 test scenes; evaluation scores all eight requests for each of these producers, giving 48 observed producer/request contexts.

Table 3: Capture-to-CAD requirements. GSD is in mm/pixel; blur and residual are in pixels; coverage is in percent; mapping error is in mm. A dash leaves the quantity unconstrained; + identifies a relaxed request.

Request	GSD $\leq$	Blur $\leq$	Coverage $\geq$	Residual $\leq$	Mapping $\leq$
Optical	1.2	3	—	—	—
Optical+	1.5	4	—	—	—
Acquisition	1	2	80	—	—
Acquisition+	1.2	3	60	—	—
Association	—	—	80	8	8
Association+	—	—	60	12	12
Complete	1	2	80	8	8
Complete+	1.2	3	60	12	12

Local dark contrast and connected components of 5–240 pixels supply candidate centers. A 30-pixel prior gate constrains CAD assignment; six or more matches trigger pose refinement with scales of 0.5 pixel, 3 mm and 0.001 rad. The implementation uses SciPy's trust-region reflective least-squares solver [30] with the specified Huber loss. The numerical dependency graph separates geometry, radiance, sensor formation, extraction, pose and optical quantities.

5.2. One continuing change campaign and matched methods

Each test source follows 16 operations: initial service, request relaxation, exposure update, request tightening, illumination update, extraction update, combined exposure/illumination/extraction update, combined camera update, stale attachment, out-of-order repair, valid repair, repair replay, numerical rollback, population mismatch, late old-version delivery and valid rollback repair. The illumination step requests the optical event; other ordinary steps request strict or relaxed complete service. The population-mismatch step executes a paired stress input with 0.14 missingness, 0–35% occlusion and larger CAD displacements. Its registered population differs from the calibrated population.

Whole renewal recomputes the complete numerical packet and constructs exact-event evidence. The primary baseline uses a persistent dependency graph, exact-event cards and the authoritative registry. It reuses an exact card only when the event and its producer projection match; containment additionally permits a stronger event to cover a weaker request. Containment with repair uses the same graph and registry with stronger-event reuse and exact fallback. Independent rules implement the same event and binding conditions. All methods receive identical inputs, trusted outputs and calibration records. Stale delivery and repair operate on the actual stored numerical packets, including their image and correspondence hashes. The independent checker reconstructs the required revision, binding and admission for every operation.

A complementary component-registration experiment uses 64 additional independent scenes. It compares clean updates with three conditional hidden couplings: image region to focal length, illumination to exposure, and extraction to focal length. Every pair re-executes the affected numerical producer; changes beyond 10^{-10} absolute/relative tolerance veto a declared invariance. This experiment tests the dependency interface before the sequential campaign applies the registered configurations.

Evidence preparation has two demand schedules: all 64 fixed contexts and the requests visited by the 16-step campaign. Each schedule builds one population-level catalogue per method, shared by all test sources. We measure its constructed cards, calibration-record evaluations and process CPU. Numerical workflow timing separately measures the 64 paired trajectories, with method order rotated across sources. The complete packet is requested at each numerical step; stage caching and evidence reuse therefore have distinct work counters. All unsuccessful cards and refused requests remain in the measurements.

The experiment runs on a shared Linux AMD EPYC 9V74 host, Python 3.12, with two workers and one numerical thread per worker. Process CPU and elapsed time include executed numerical stages, geometric quality scoring, payload hashing and registry/consistency checks. Initial nominal-scene sampling, process creation and file export are excluded; the stress-input draw is included in its population-change step. Catalogue preparation is timed separately. A 10,000-replicate paired bootstrap resamples the 64 complete sources for the workflow CPU comparison. The Supplementary Material records the environment, all event outcomes and the earlier single-update and repair studies.

5.3. Surface-structure and imaging challenge

The geometric challenge uses Blender 4.3.2 Cycles to evaluate light transport through explicit geometry, materials and illumination [31]. Its four structural families are a longitudinal lap joint, rounded access hatch, doubly curved diagonal joint and raised repair doubler. Individual scenes sample part dimensions, joint or panel contours, fastener positions, curvature, materials and camera pose. Geometric sealant, shallow fastener heads, painted and metallic surfaces, abrasion and three area emitters produce the image. Figure 3 shows fixed-source examples of the processed images.

Eight development scenes cover the first two families. The test set has 16 independent scenes per family, giving 64 scenes and 256 images under nominal, raking-light, oblique-view and foreground-occlusion conditions. The latter two families are held out from image-provider selection. Conditions change actual emitters, camera pose or foreground hose/tool geometry. Resolution is 768×576, with at most 32 samples per pixel, adaptive threshold 0.02 and no denoising. Camera projection and skin-mesh ray intersection are checked independently before scoring.

The original contrast extractor is compared with a locally normalized extractor. For grayscale intensity I on the 0–255 scale, the alternative uses contrast $c = G_3 * I - I$ and local scale $s = [\max\{G_8 * I^2 - (G_8 * I)^2, 0\}]^{1/2}$, where G_σ is a Gaussian filter. A candidate pixel satisfies both $c > a$ and $c > z \max(s, 1)$. Development selects one of the six predefined pairs $z \in \{1.0, 1.6, 2.2\}$ and $a \in \{4, 8\}$, minimizing quality failures and then maximizing coverage across equally weighted family/condition cells. Both providers use the same component filters, assignment and pose estimator. Test scoring begins after this selection is fixed.

The challenge applies the Association thresholds in Table 3: correct-address coverage ≥ 0.8, maximum fitted residual ≤ 8 pixels and maximum all-site mapping error ≤ 8 mm. Correct-address scoring additionally requires a visible, nonmissing target with the assigned CAD identity and image error below 8 pixels. Missing and occluded sites remain in the denominator; failed surface intersections remain as infinite mapping errors. The observable screen instead checks matched-site fraction ≥ 0.8 and fitted residual ≤ 8 pixels. The two fixed providers produce 512 evaluated outputs. We report quality failures, screen acceptance and accepted-but-failed outputs for every family and condition. A paired bootstrap with 4,000 replicates resamples scenes within each family, retaining all repeated conditions and both providers. These images define a separate evaluated population with their own recorded quality outcomes.

6. Results

6.1. The supported service changes with the requirement

Calibration supports 54 of the 64 producer/request contexts (Fig. 4). All four complete methods retain this same set. Under the fixed catalogue order, containment supplies 29 contexts from existing stronger cards; exact construction supports 25 further contexts and refuses ten. The exposure update raises the strict complete-event bound to 0.292, while its relaxed event remains at 0.104. The combined exposure, illumination and extraction update similarly supports relaxed complete service at 0.130 while refusing strict

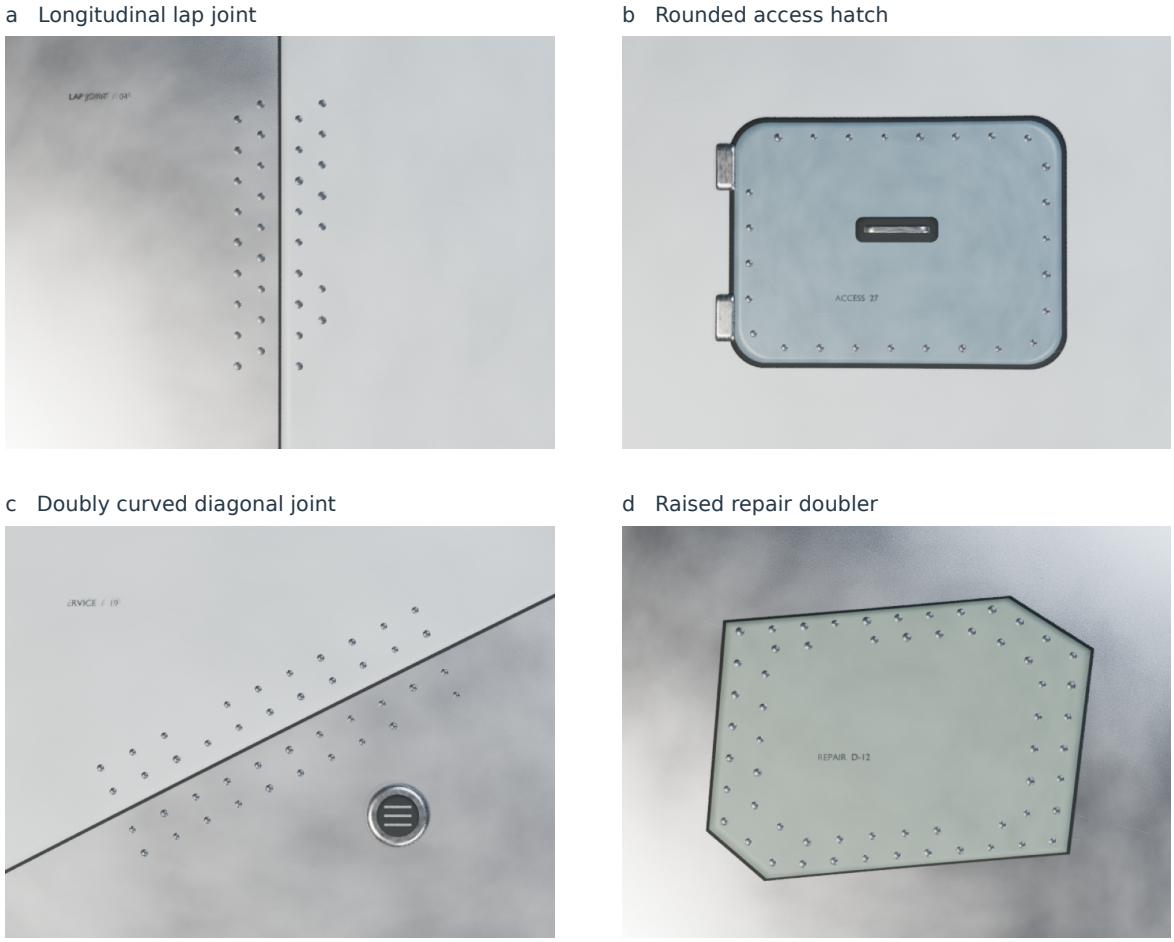


Figure 3: Actual nominal inputs from the surface challenge. Mesh topology changes across the four families, and dimensions, fastener layouts, surface appearance and camera pose vary within each family. The first test source from each family is shown without image retouching. Diagonal-joint and repair-doubler structures are held out from image-provider selection.

complete service at 0.352. Under reduced illumination, the optical event remains supported at 0.028; complete association requirements determine the additional refusals.

The six producers used in the sequential test yield 48 observed contexts, of which 38 were admitted by calibration. Among these 38 calibration-admitted contexts, test failures range from 0 to 4 per 64 scenes. At baseline, mean correct-address coverage is 95.94%, median fitted residual is 0.646 pixel, and median mapping error is 0.586 mm. Median prior translation error decreases from 4.431 to 1.254 mm after fitting. All test outcomes, including the refused programs and individual quality failures, are retained in the Supplementary Material.

In the separate 64-source component-registration experiment (Supplementary Material), paired numerical re-execution detects all 86 activated hidden couplings, with no veto among its 192 clean update pairs. These checks exercise numerical effects of a violated dependency declaration. The sequential experiment then examines the evidence and output history of the registered updates.

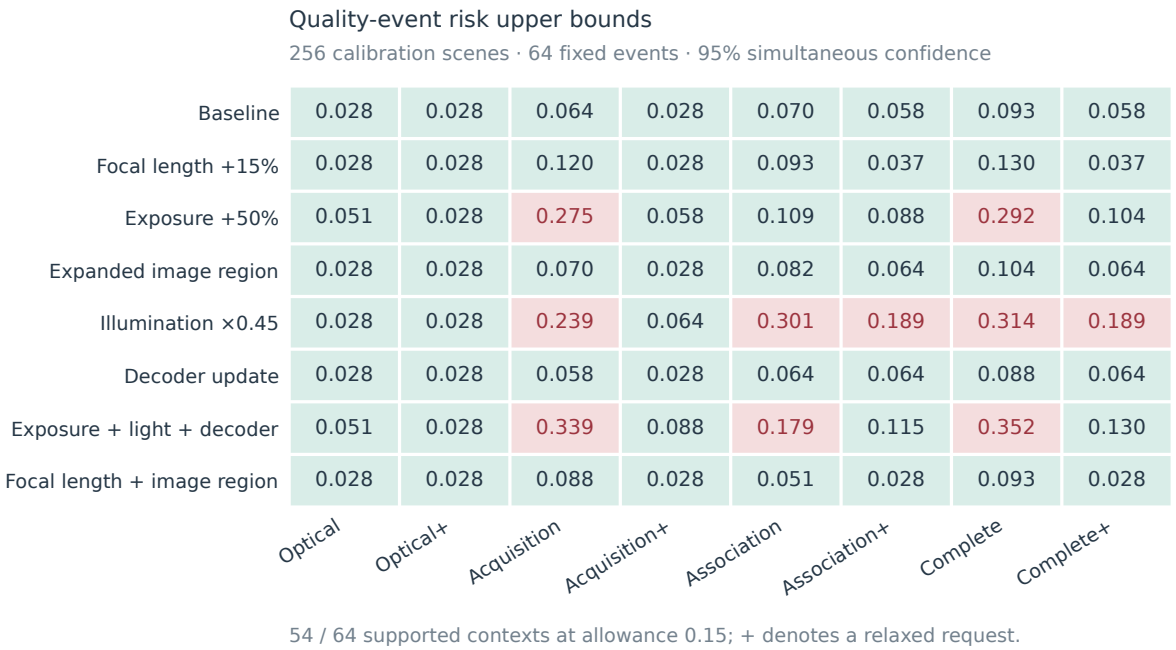


Figure 4: Exact-event calibration bounds for the 64 fixed contexts. Green cells meet the 0.15 scene-level allowance. The 256 calibration scenes are independent of the 64 campaign test sources. The three incremental methods and whole renewal support the same 54 contexts.

6.2. Continuing service through repair and rollback

The campaign distinguishes three causes of refusal within the same output history (Fig. 5). Tightening the exposure-updated request fails its risk allowance despite a current authenticated binding. Stale delivery and an out-of-order repair encounter an admissible event but fail the binding rules. The stress input has a different registered population, so its quality statement is unavailable before any attachment decision. These decisions follow from the event, producer and output records supplied to each method.

All four methods make the same 704 issuance decisions among 1,024 operations per implementation. No stale, wrong-origin or invalid-revision binding is issued. Both authorized repairs succeed for each of the 64 sources, restoring all 128 eligible recovery checkpoints. A replayed repair is rejected while the already repaired output remains available. Numerical rollback restores the cached baseline output at registry revision 9, advancing from revision 8; the incremental methods execute no numerical provider for that operation. A late delivery carrying the earlier revision is still refused, and the final repair restores service for all 64 sources. Thus reuse of numerical data and reuse of control authority follow different revision rules.

6.3. Evidence work and numerical computation

For the complete family, whole renewal constructs 64 cards, exact dependency caching constructs 54, and containment with fallback constructs 35. The corresponding record evaluations are 16,384, 13,824 and 8,960. Containment reduces this work by 35.19% relative to exact caching while preserving all 54 supported contexts. The independent rules give the same construction and support counts. For the requests actually visited in the 16-step campaign, exact caching constructs eight cards and containment constructs six: 2,048 versus 1,536 record evaluations, a 25% reduction. Each catalogue is prepared once and shared by all 64 sources.

Figure 6 separates this evidence work from numerical execution. Whole renewal consumes 193.617 s of process CPU across the 64 trajectories, compared with 78.773 s for the DAG baseline. The shared dependency cache reduces workflow CPU by 59.31%, with a paired 95% bootstrap interval of 58.50–60.12%. Containment and rules consume 79.328 and 79.687 s, with identical numerical outputs and stage counts. A separate cold

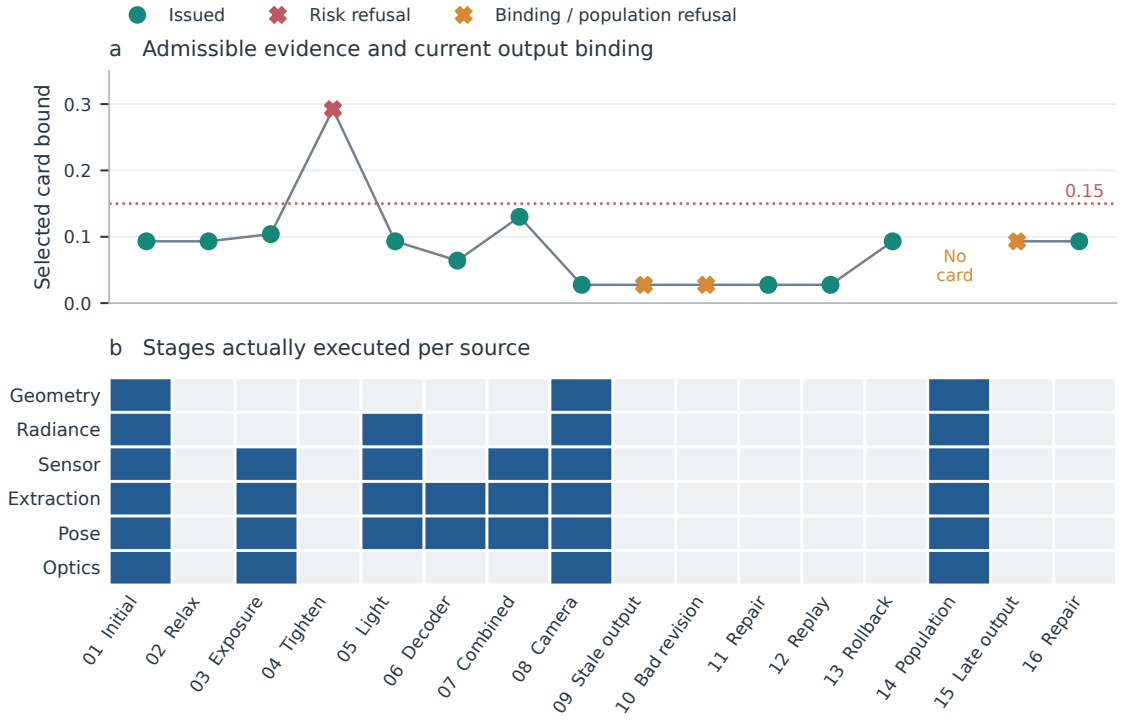


Figure 5: The 16-operation lifecycle. (a) The selected evidence bound and actual issuance decision. Reused stronger cards can have a higher bound than the request's exact card. A statistically supported service can still be withheld for a stale binding; a population mismatch has no applicable card. (b) Actual dependency-stage execution. Valid repair and numerical rollback use retained outputs. The independent registry and rule implementations reproduce these decisions and stage counts.

full-catalogue preparation takes 9.577 ms for exact caching and 6.324 ms for containment. The principal operational result is selective continuation and recovery with explicit evidence applicability; the additional containment saving concerns this small preparation task.

6.4. Quality under structural and imaging changes

Development selects the normalized extractor with $z = 1$ and $a = 8$. Figure 7 retains all 256 test images and compares this fixed choice with the original extractor. In nominal, raking-light, oblique-view and occluded conditions, the original extractor fails the association event on 9, 27, 11, 35 of 64 scenes, respectively. The normalized extractor gives 22, 26, 18, 38 failures under the same conditions. Its mean correct-address coverage is 96.3, 89.7, 95.7, 87.8%, respectively. The paired coverage differences and their scene-bootstrap intervals show the effect of replacing this component under each condition. Across all conditions, the development-selected extractor has 104 event failures, compared with 82 for the original extractor. Under nominal imaging, mean coverage increases from 95.00% to 96.27%, while residual violations increase from 9 to 22 of 64 scenes. The coverage gain is accompanied by more residual failures, making the complete quality event relevant to component replacement.

The two structurally held-out families contribute 32 independent scenes and 128 condition images. The normalized extractor has 62 quality failures among these images, compared with 42 among the 128 images from familiar structural families. Within the paired occlusion intervention, 18 scenes change from a nominal pass to failure and 2 change from failure to pass. Mean coverage changes by -8.50 percentage points, with a

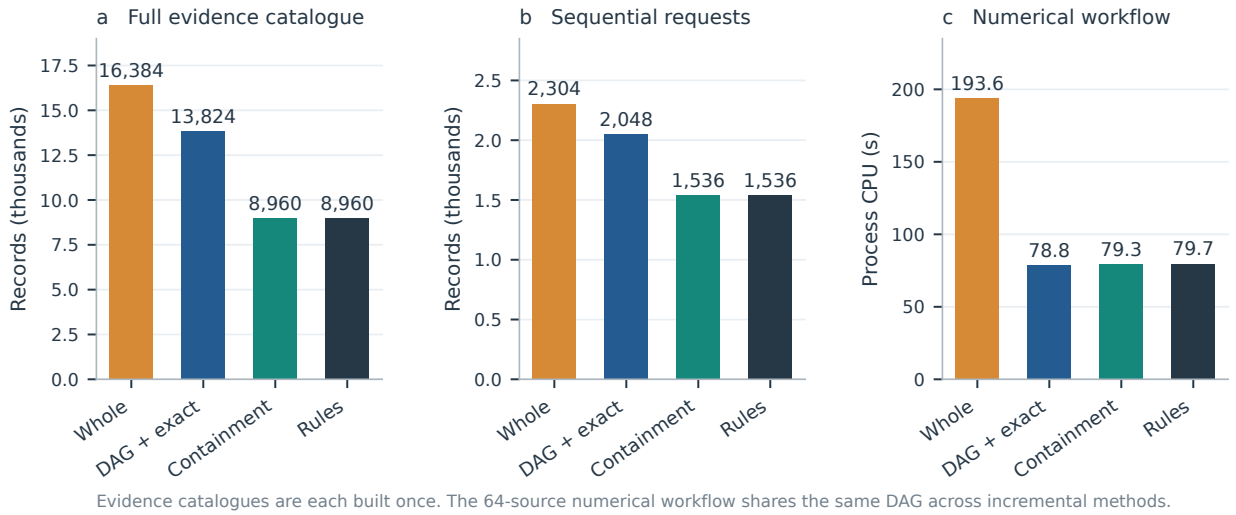


Figure 6: Attribution of computational work. The complete family and the sequential demand each build one shared evidence catalogue. Event containment reduces record evaluations relative to exact caching. The larger workflow CPU saving comes from the ordinary dependency graph shared by the three incremental methods.

paired 95% interval of -9.41 to -7.66. Planned targets remain unchanged across each pair, so occluded or missing sites still contribute to the requested coverage.

The observable matched-count/residual screen accepts 174 of 256 outputs from the original extractor and 152 from the normalized extractor. Of these, 0 and 0 outputs, respectively, fail the independently scored quality event. Figure 8 shows the first repair-doubler source before and after foreground occlusion, with fitted CAD projections and pixel-derived centers. Correct matches decrease from 59 to 51 of 61 planned sites. The occluded output retains 4 incorrect assignments; its maximum residual rises from 5.05 to 24.62 pixels, crossing the 8-pixel requirement, while mapping error remains below 8 mm. These measurements distinguish a current, authenticated numerical result from the geometric quality of the result. The challenge outcomes remain associated with this renderer and image provider when the evidence interface evaluates applicability.

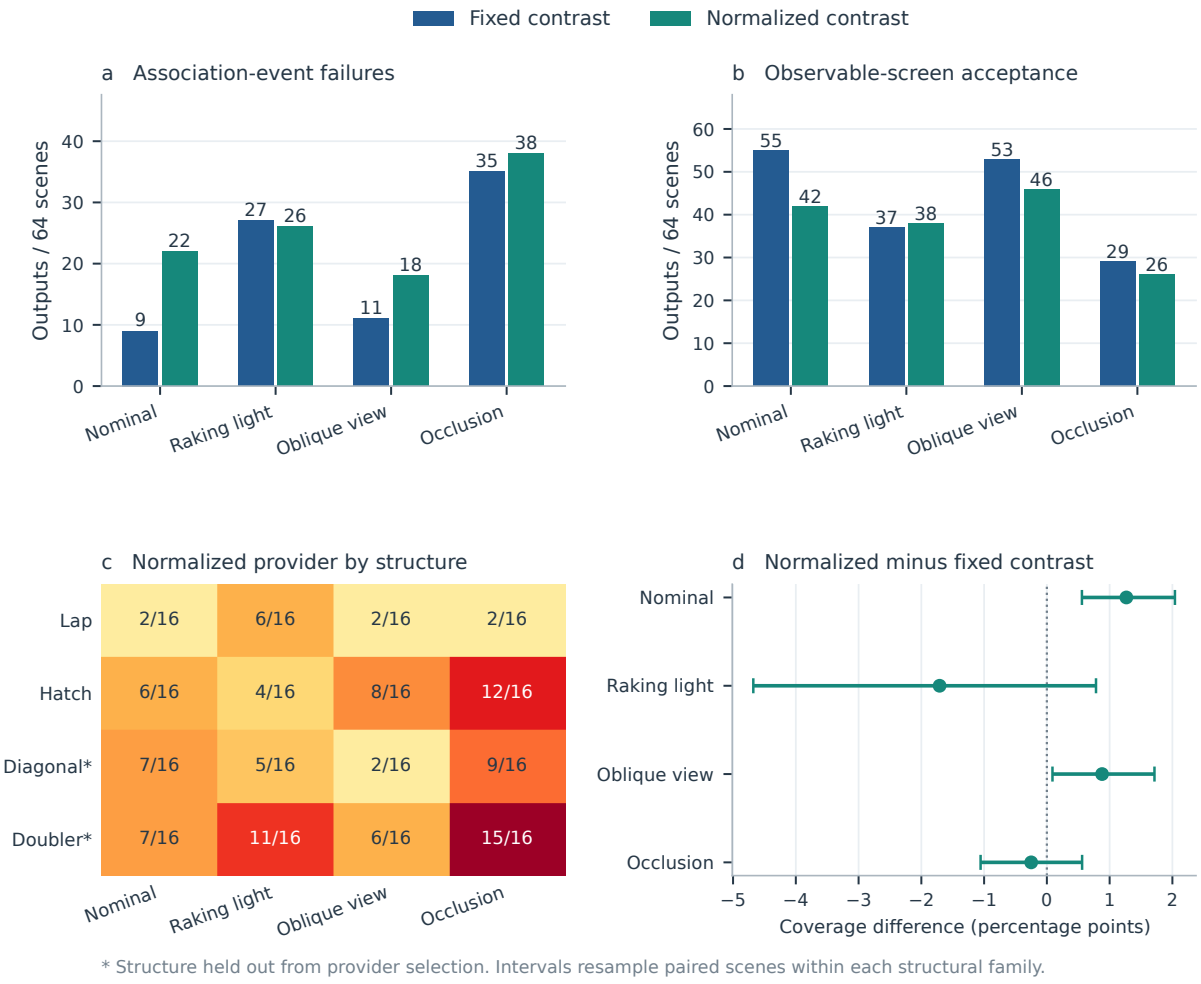


Figure 7: Complete test outcomes on 64 independent scenes under four conditions. (a) Violation of coverage, fitted-residual or mapping requirements. (b) Outputs accepted by the observable count/residual screen; none of these accepted outputs fails the independently scored association event. (c) Failures of the development-selected provider within each structural family. (d) Paired mean coverage differences with descriptive 95% bootstrap intervals, resampling scenes within families.

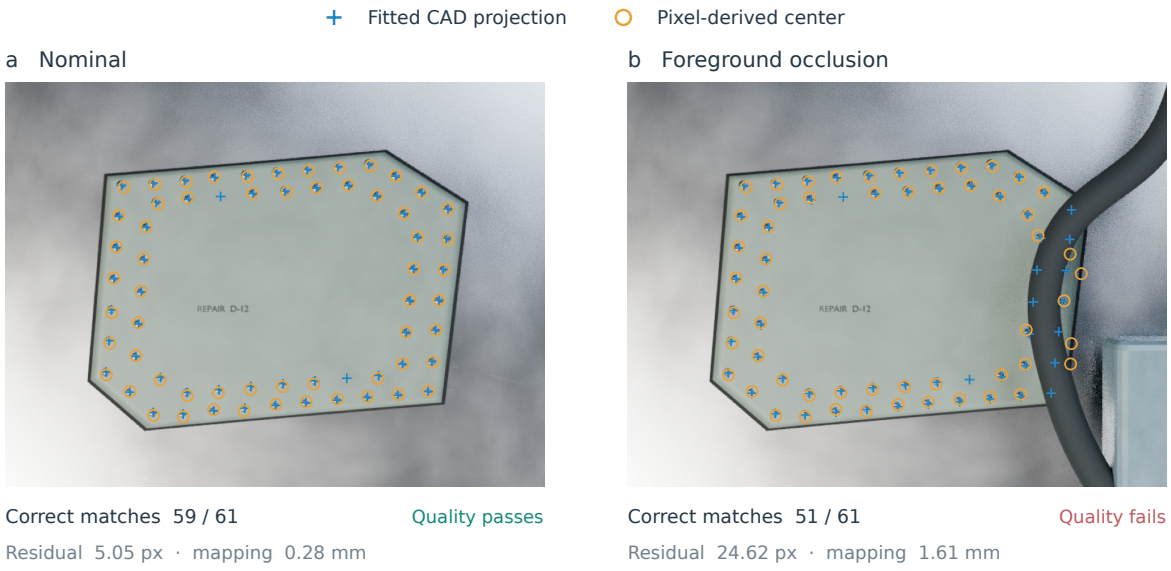


Figure 8: Actual correspondence outputs for a fixed test pair, using the development-selected normalized extractor. Both frames are unretouched renderer outputs. Overlays show all fitted CAD locations and the assigned pixel centers; the scoring denominator contains the same planned sites in both conditions. This source was selected by its position in the fixed test list.

7. Discussion

The proposed interface maintains the applicability of a quantitative inspection statement through engineering change. Lifecycle knowledge graphs preserve the relationships among identities, specifications and operating states [6], and dynamic safety cases update evidence as monitored conditions change [7]. Here the maintained object is a calibrated quality event attached to a particular output. Requirement containment, producer compatibility and the current registry binding jointly determine whether the service is retained, renewed, refused or recovered. The sequential campaign places these decisions within one output history, including combined producer updates, repair replay and a numerical rollback at a newer registry revision.

The matched comparisons identify the role of each mechanism. Dependency caching retains unaffected numerical stages, following selective recomputation [15, 17]. Event containment reduces the records needed to prepare a quality statement, while exact fallback preserves the admission decision of the complete catalogue. Registry transactions determine which numerical output can carry that statement. The separate work measurements make these effects assessable under the same producer, calibration inputs and authoritative records. Independent rule and registry implementations test whether the decisions follow from the specified semantics.

Physical quality and output integrity have complementary roles. Correcting a stale attachment restores the association to a trusted output; the event's calibrated bound determines whether the requested service can resume. An output with acceptable fitted residual can still contain incorrect CAD associations, which motivates evaluation against all requested quantities and the challenge's accepted-but-failed measure. The four surface families and four imaging conditions examine this distinction with the same planned sites retained under changes in appearance, view and occlusion. Provider selection uses development scenes; the structure-held-out families evaluate its behavior under new surface organization.

The workflow operates with declared provider dependencies and a trusted registry. Functional identity supplies the transfer condition, and paired regression checks its implementation on executed inputs. Population identity keeps each calibrated statement attached to its operating distribution. These explicit conditions make a renewed or repaired output inspectable: its record identifies the requested quality, the producer establishing that quality and the current output carrying the evidence.

8. Conclusion

This study develops an executable interface for carrying quantitative inspection evidence through changes in requirements, numerical producers and output versions. Event containment reuses a stronger quality statement; exact fallback preserves the support of the complete calibrated catalogue. An authenticated repair transaction then restores the output that can carry the statement, with numerical rollback and registry revision treated separately.

Complete-family evaluation retains all 54 calibrated producer/request contexts while reducing full-catalogue evidence evaluations by 35.19% relative to exact dependency caching. The separate sequential-demand catalogue reduces evaluations by 25%. All complete implementations restore their 128 eligible repair checkpoints, with no invalid binding issued across the 1,024 operations of each implementation. The geometric image challenge quantifies the effects of surface structure, view and occlusion on association quality, linking service requirements to actual correspondence and mapping outcomes. Together, these results establish a reproducible decision process for retaining applicable evidence, renewing affected statements and resuming an inspection service on the correct output.

References

[1] I. Jovančević, I. Viana, J.-J. Orteu, T. Sentenac, S. Larnier, Matching CAD model and image features for robot navigation and inspection of an aircraft, in: Proceedings of the 5th International Conference on Pattern Recognition Applications and Methods, 2016, pp. 359–366. doi:10.5220/0005756303590366.

[2] O. S. Amosov, S. G. Amosova, I. O. Iochkov, Deep neural network recognition of rivet joint defects in aircraft products, Sensors 22 (9) (2022) 3417. doi:10.3390/s22093417.

[3] S. Ma, K. A. Flanigan, M. Bergés, State-of-the-art review and synthesis: A requirement-based roadmap for standardized predictive maintenance automation using digital twin technologies, Advanced Engineering Informatics 62 (2024) 102800. doi:10.1016/j.aei.2024.102800.

[4] K. Hu, C. Yan, C. Zhou, T. Deng, L. Wang, Y. Xu, C. Zhuang, Adaptive threshold optimization for efficient and robust high-dimensional sensor calibration in building energy systems, Advanced Engineering Informatics 77 (2027) 105365, available online 10 October 2026. doi:10.1016/j.aei.2026.105365.

[5] Q. Zou, H.-Y. Feng, A decision-support method for information inconsistency resolution in direct modeling of CAD models, Advanced Engineering Informatics 44 (2020) 101087. doi:10.1016/j.aei.2020.101087.

[6] C. Yang, Y. Liu, H. Dong, P. Zhang, J. Lu, B. Zhu, R. Ojala, K. Tammi, Lifecycle-aware knowledge graphs with human-in-the-loop agentic retrieval for automotive manufacturing, Advanced Engineering Informatics 76 (2026) 105122. doi:10.1016/j.aei.2026.105122.

[7] E. Denney, G. Pai, I. Habli, Dynamic safety cases for through-life safety assurance, in: 2015 IEEE/ACM 37th IEEE International Conference on Software Engineering, IEEE, 2015, pp. 587–590. doi:10.1109/ICSE.2015.199.

[8] L. Moreau, P. Missier, PROV-DM: The PROV data model, W3C recommendation, World Wide Web Consortium, editors; Recommendation of 30 April 2013 (Apr. 2013).
URL https://www.w3.org/TR/2013/REC-prov-dm-20130430/

[9] L. Petnga, M. Austin, An ontological framework for knowledge modeling and decision support in cyber-physical systems, Advanced Engineering Informatics 30 (1) (2016) 77–94. doi:10.1016/j.aei.2015.12.003.

[10] C. Quinn, J. J. McArthur, A case study comparing the completeness and expressiveness of two industry recognized ontologies, Advanced Engineering Informatics 47 (2021) 101233. doi:10.1016/j.aei.2020.101233.

[11] L. Ma, T. Hartmann, A proposed ontology to support the hardware design of building inspection robot systems, Advanced Engineering Informatics 55 (2023) 101851. doi:10.1016/j.aei.2022.101851.

[12] J. Xiao, Q. Wang, Y. Li, Z. Yu, H. Zhou, B. Lin, A fully automated DM-BIM-BEM pipeline enabling graph-based intelligence, interoperability, and performance-driven early design, Advanced Engineering Informatics 75 (2026) 104824. doi:10.1016/j.aei.2026.104824.

[13] J. Cheney, P. Missier, L. Moreau, Constraints of the PROV data model, W3C recommendation, World Wide Web Consortium, editors; Recommendation of 30 April 2013; contributing author Tom De Nies (Apr. 2013).
URL https://www.w3.org/TR/2013/REC-prov-constraints-20130430/

[14] P. Missier, S. Woodman, H. Hiden, P. Watson, Provenance and data differencing for workflow reproducibility analysis, Concurrency and Computation: Practice and Experience 28 (4) (2016) 995–1015. doi:10.1002/cpe.3035.

[15] P. Missier, J. Cała, M. Rathi, Preserving the value of large scale data analytics over time through selective re-computation, in: Data Analytics, Vol. 10365 of Lecture Notes in Computer Science, Springer, 2017, pp. 65–77. doi:10.1007/978-3-319-60795-5_6.

[16] J. Cała, P. Missier, Provenance annotation and analysis to support process re-computation, in: Provenance and Annotation of Data and Processes, Vol. 11017 of Lecture Notes in Computer Science, Springer, 2018, pp. 3–15. doi:10.1007/978-3-319-98379-0_1.

[17] A. Mokhov, N. Mitchell, S. Peyton Jones, Build systems à la carte, Proceedings of the ACM on Programming Languages 2 (ICFP) (2018) 1–29. doi:10.1145/3236774.

[18] S. Martínez-Sanchis, SafeMachine: An evidence-centric architecture for traceable and context-aware machine safety assessment, Advanced Engineering Informatics 77 (2027) 105321, available online in 2026. doi:10.1016/j.aei.2026.105321.

[19] A. Benveniste, B. Caillaud, D. Nickovic, R. Passerone, J.-B. Raclet, P. Reinkemeier, A. Sangiovanni-Vincentelli, W. Damm, T. A. Henzinger, K. G. Larsen, Contracts for system design, Foundations and Trends in Electronic Design Automation 12 (2–3) (2018) 124–400. doi:10.1561/1000000053.

[20] U. Mehmood, S. Sheikhi, S. Bak, S. A. Smolka, S. D. Stoller, The black-box simplex architecture for runtime assurance of autonomous CPS, in: NASA Formal Methods, Vol. 13260 of Lecture Notes in Computer Science, Springer, 2022, pp. 231–250. doi:10.1007/978-3-031-06773-0_12.

[21] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, The Annals of Applied Statistics 19 (2) (2025) 1641–1662. doi:10.1214/24-AOAS1998.

[22] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, T. Schuster, Conformal risk control, in: The Twelfth International Conference on Learning Representations, 2024.
URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/f3549ef9b5ff520a7e41ff3cc306ab2b-Abstract-Conference.html

[23] C. J. Clopper, E. S. Pearson, The use of confidence or fiducial limits illustrated in the case of the binomial, Biometrika 26 (4) (1934) 404–413. doi:10.1093/biomet/26.4.404.

[24] T. Liu, B. Li, X. Du, B. Jiang, X. Jin, L. Jin, Z. Zhao, Component-aware anomaly detection framework for adjustable and logical industrial visual inspection, Advanced Engineering Informatics 58 (2023) 102161. doi:10.1016/j.aei.2023.102161.

[25] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, P. Abbeel, Domain randomization for transferring deep neural networks from simulation to the real world, in: 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017, pp. 23–30. doi:10.1109/iros.2017.8202133.

[26] C. Xu, G. Vu, B. T. Cao, Z. Liu, F. Diewald, Y. Yuan, G. Meschke, Bridging simulation and experiment: A self-supervised domain adaptation framework for concrete damage classification, Advanced Engineering Informatics 69 (2026) 104125. doi:10.1016/j.aei.2025.104125.

[27] R. Hartley, A. Zisserman, Multiple View Geometry in Computer Vision, 2nd Edition, Cambridge University Press, 2004. doi:10.1017/CBO9780511811685.

[28] D. F. Crouse, On implementing 2D rectangular assignment algorithms, IEEE Transactions on Aerospace and Electronic Systems 52 (4) (2016) 1679–1696. doi:10.1109/TAES.2016.140952.

[29] P. J. Huber, Robust estimation of a location parameter, The Annals of Mathematical Statistics 35 (1) (1964) 73–101. doi:10.1214/aoms/1177703732.

[30] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, SciPy 1.0 Contributors, SciPy 1.0:

fundamental algorithms for scientific computing in Python, Nature Methods 17 (3) (2020) 261–272. doi:10.1038/s41592-019-0686-2.

[31] J. T. Kajiya, The rendering equation, ACM SIGGRAPH Computer Graphics 20 (4) (1986) 143–150. doi:10.1145/15886.15902.